THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 5: **Prime Time** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of all distinct prime factors of 390 is: 1

(a) 26 **(b)** 20 **(c)** 390 **(d)** 23

2. The number of factors of 100 is: 1

(a) 7 **(b)** 9 **(c)** 11 **(d)** 4

3. The HCF of 63 and 91 is: 1

(a) 7 **(b)** 14 **(c)** 21 **(d)** 819

4. The LCM of 14 and 35 is: 1

(a) 140 **(b)** 490 **(c)** 70 **(d)** 35

5. The smallest digit to put in the blank to make 364_ divisible by 3 is: 1

(a) 4 **(b)** 2 **(c)** 6 **(d)** 3

6. The largest 3-digit multiple of 23 is: 1

(a) 999 **(b)** 943 **(c)** 966 **(d)** 989

7. How many prime numbers are there between 70 and 110? 1

(a) 10 **(b)** 12 **(c)** 9 **(d)** 11

8. The sum of all distinct prime factors of 462 is: 1

(a) 462 **(b)** 26 **(c)** 20 **(d)** 23

9. The number of factors of 120 is: 1

(a) 16 **(b)** 3 **(c)** 5 **(d)** 18

10. The HCF of 54 and 90 is: 1

(a) 270 **(b)** 54 **(c)** 36 **(d)** 18

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the missing digit so that 619_ is divisible by 9. 2

12. Find the HCF and LCM of 48 and 72 using prime factorisation. 2

13. Find the smallest number that must be added to 7891 to make it divisible by 8. 2

14. Find the missing digit so that 904_ is divisible by 9. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the prime factorisation of 392 and the number of its factors. 3

16. Two bells ring every 10 minutes and every 25 minutes. They ring together at 9 am. When do they next ring together? 3

17. Two ropes of length 75 m and 100 m are cut into pieces of equal length, as long as possible, with nothing left over. Find the length of each piece and the total number of pieces. 3

OR

Find the prime factorisation of 450 and the number of its factors. 3

18. Two bells ring every 16 minutes and every 20 minutes. They ring together at 9 am. When do they next ring together? 3

OR

Two ropes of length 105 m and 135 m are cut into pieces of equal length, as long as possible, with nothing left over. Find the length of each piece and the total number of pieces. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A room is 720 cm long and 540 cm wide. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

OR

Find the smallest number that leaves a remainder 1 when divided by 4, 6 and 9, and the next such number. 5

20. A room is 840 cm long and 630 cm wide. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

OR

Find the smallest number that leaves a remainder 1 when divided by 3, 5 and 9, and the next such number. 5

END OF QUESTION PAPER — Total: 40 Marks

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